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4. Information Gathering: Interactive Methods

System Analysis and Design

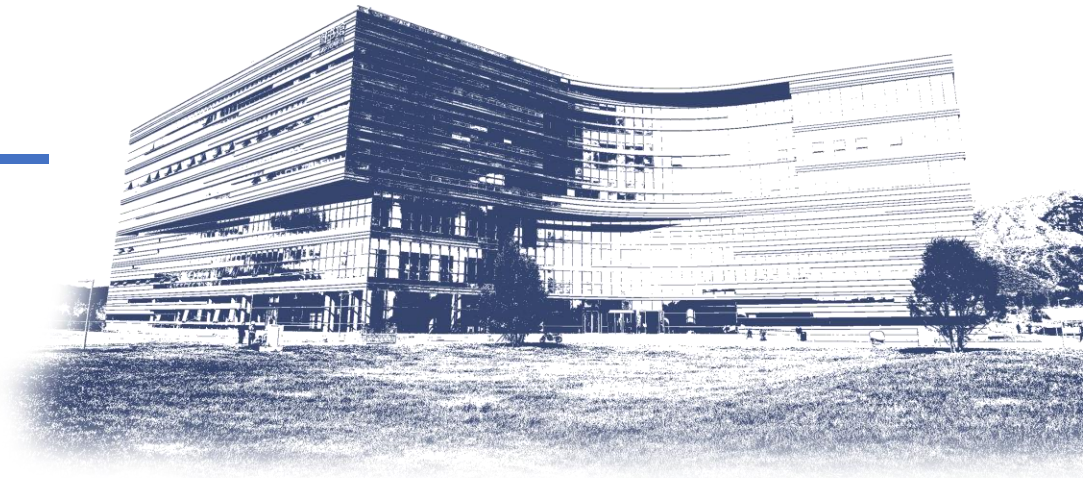
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Learning Objectives

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- Recognize the value of interactive methods for information gathering
- Construct interview questions to elicit human information requirements and structure them in a way that is meaningful to users
- Understand the purpose of stories and why they are useful in systems analysis
- Understand the concept of JAD and when to use it
- Write effective questions to survey users about their work
- Design and administer effective questionnaires

Interactive Methods to Elicit Human Information Requirements

- Interviewing
- Joint Application Design (JAD)
- Questionnaires

Major Topics (1 of 2)

- Interviewing
  - Interview preparation
  - Question types
  - Arranging questions
  - The interview report
- User Stories

Major Topics (2 of 2)

- Joint Application Design (JAD)
  - Involvement
  - Location
- Questionnaires
  - Writing questions
  - Using scales
  - Design
  - Administering

4.1 Interviewing

- Interviewing is an important method for collecting data on human and system information requirements
- Interviews reveal information about:
  - Interviewee opinions
  - Interviewee feelings
  - Goals
  - Key HCI concerns

4.1.1 Interview Preparation

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure

4.1.2 Question Types (1 of 2)

- **Open-ended**
- **Closed**

Open-Ended Questions

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- Open-ended interview questions allow interviewees to respond how they wish without length or structure limitations
- Open-ended interview questions are appropriate when the analyst is interested in breadth and depth of reply

Open-Ended Interview Questions

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- What's your opinion of the current state of business-to-business ecommerce in your firm?
- What are the critical objectives of your department?
- Once the data are submitted via the website, how are they processed?
- Describe the monitoring process that is available online.
- What are some of the common data entry errors made in this department?
- What are the biggest frustrations you've experienced during the transition to ecommerce?

Advantages of Open-Ended Questions (1 of 2)

- Puts the interviewee at ease
- Allows the interviewer to pick up on the interviewee's vocabulary
- Provides richness of detail
- Reveals avenues of further questioning that may have gone untapped

Advantages of Open-Ended Questions (2 of 2)

- Provides more interest for the interviewee
- Allows more spontaneity
- Makes phrasing easier for the interviewer
- Useful if the interviewer is unprepared

Disadvantages of Open-Ended Questions

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- May result in too much irrelevant detail
- Possibly losing control of the interview
- May take too much time for the amount of useful information gained
- Potentially seeming that the interviewer is unprepared
- Possibly giving the impression that the interviewer is on a “fishing expedition”

Closed Interview Questions (封闭式问题)

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- Closed interview questions limit the number of possible responses
- Closed interview questions are appropriate for generating precise, reliable data that is easy to analyze
- The methodology is efficient, and it requires little skill for interviewers to administer

Closed Interview Questions

- How many times a week is the project repository updated?
- On average, how many calls does the call center receive monthly?
- Which of the following sources of information is most valuable to you?
  - Completed customer complaint forms
  - Email complaints from consumers who visit the website
  - Face-to-face interaction with customers
  - Returned merchandise
- List your top two priorities for improving the technology infrastructure.
- Who receives this input?

Benefits of Closed Interview Questions

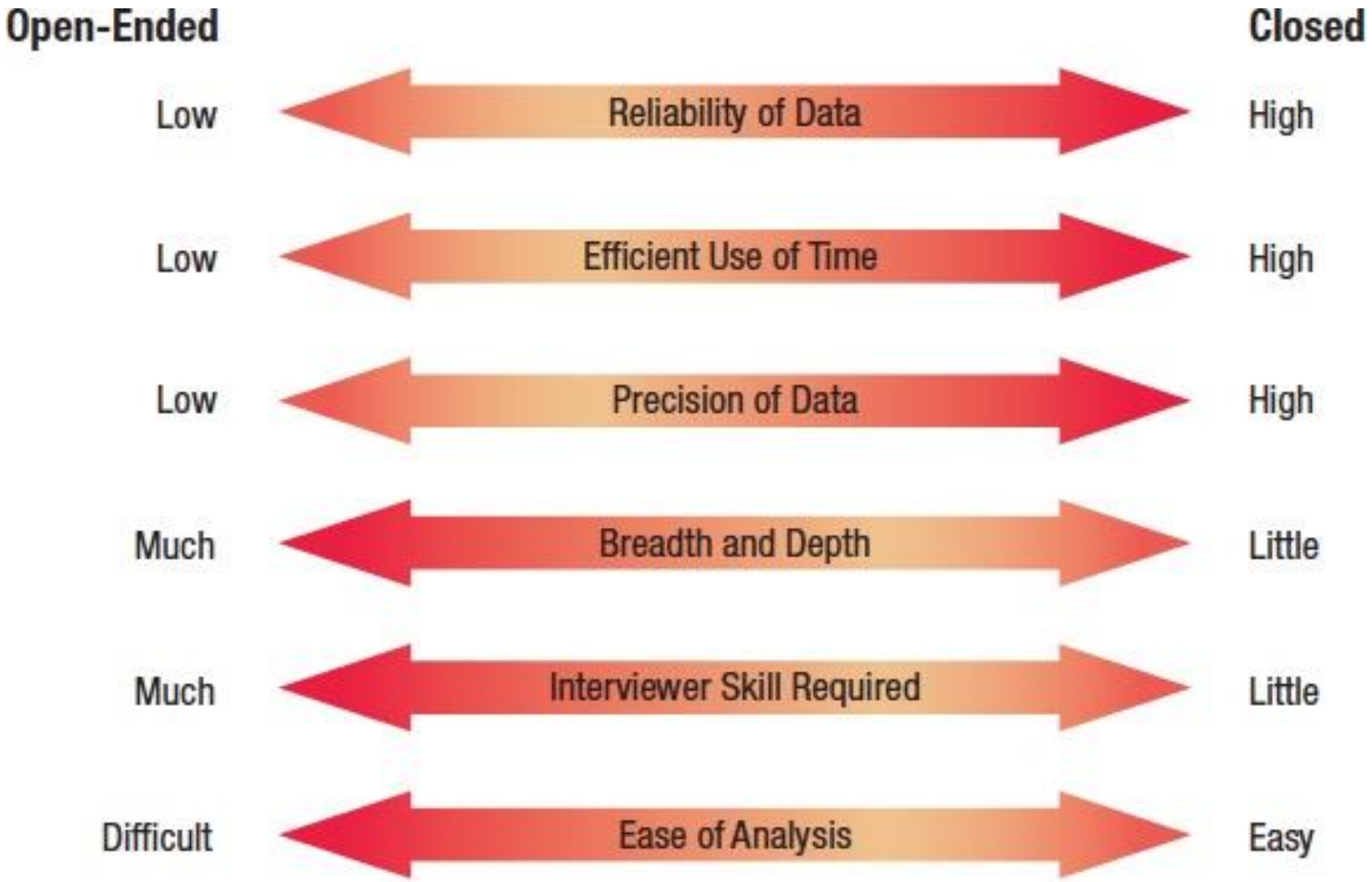
- Save interview time
- Easily compare interviews
- Quickly get to the point
- Maintain control of the interview
- Cover a large area quickly
- Obtain relevant data

Disadvantages of Closed Interview Questions

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- May be boring for the interviewee
- May fail to obtain rich details
- May miss some main ideas
- May fail to build rapport between interviewer and interviewee

Attributes of Open-Ended and Closed Questions



Bipolar Questions

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- Bipolar questions are those that may be answered with a “yes” or “no” or “agree” or “disagree”
- Bipolar questions should be used sparingly
- A special kind of closed question

Probes（调查问题）

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- Probing questions elicit more detail about previous questions
- The purpose of probing questions is:
  - To get more meaning
  - To clarify
  - To draw out and expand on the interviewee's point
- May be either open-ended or closed

4.1.3 Arranging Questions

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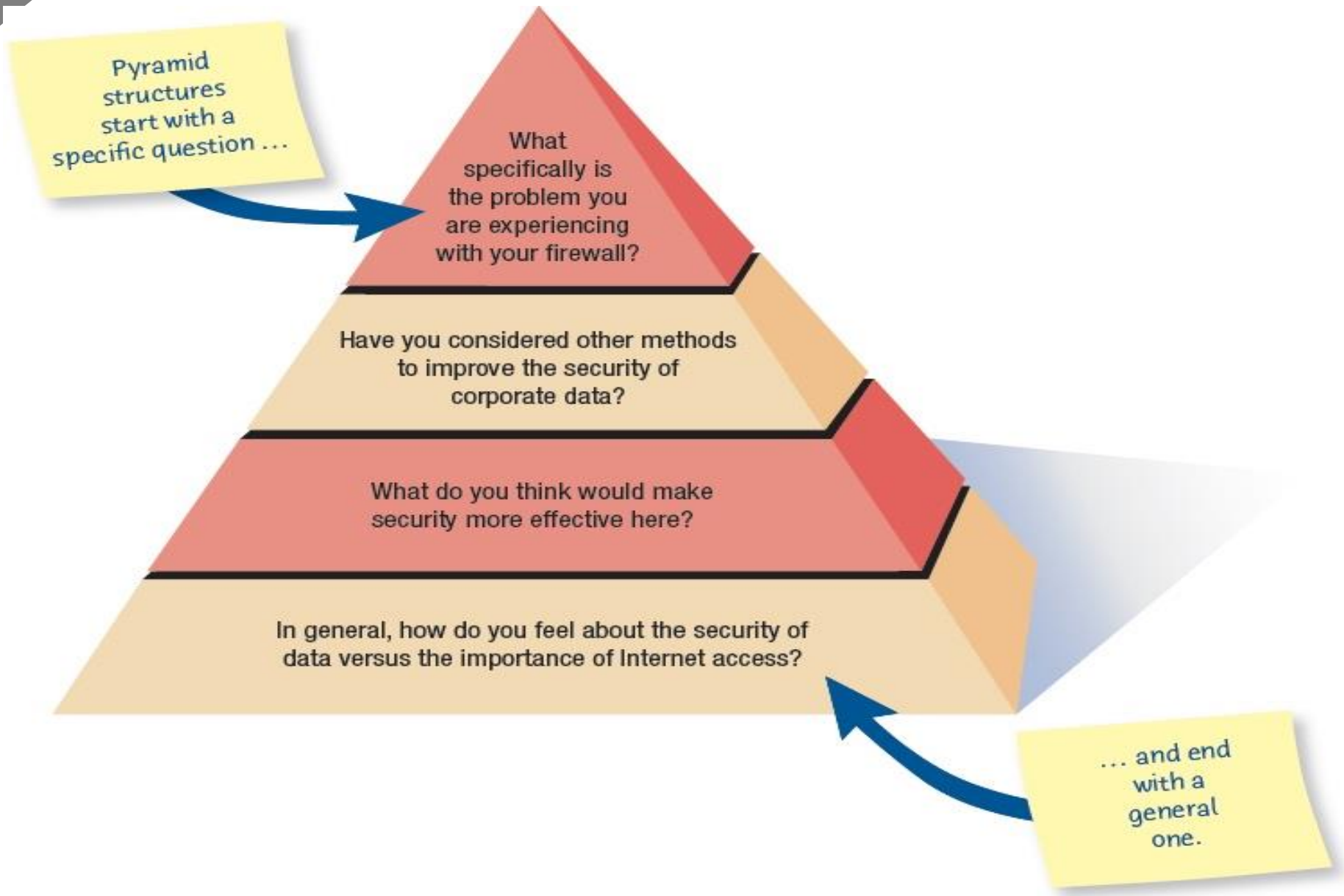
- Pyramid金字塔结构
  - Starting with closed questions and working toward open-ended questions
- Funnel漏斗结构
  - Starting with open-ended questions and working toward closed questions
- Diamond菱形结构
  - Starting with closed, moving toward open-ended, and ending with closed questions

Pyramid Structure

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- Begins with very detailed, often closed questions
- Expands by allowing open-ended questions and more generalized responses
- Is useful if interviewees need to be warmed up to the topic or seem reluctant to address the topic

Pyramid Structure for Interviewing Goes from Specific to General Questions

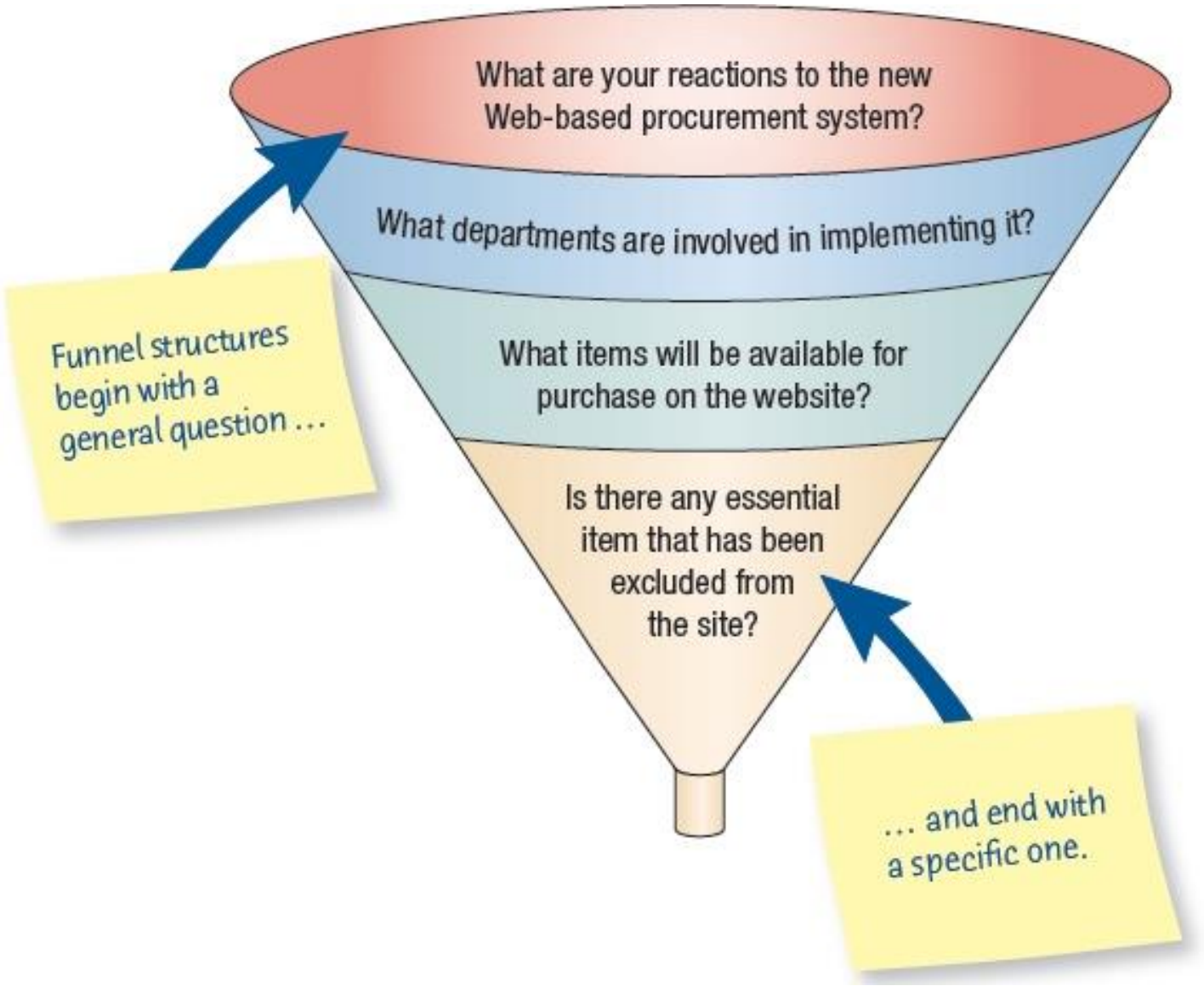


Funnel Structure

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- Begins with generalized, open-ended questions
- Concludes by narrowing the possible responses using closed questions
- Provides an easy, nonthreatening way to begin an interview
- Is useful when the interviewee feels emotionally about the topic

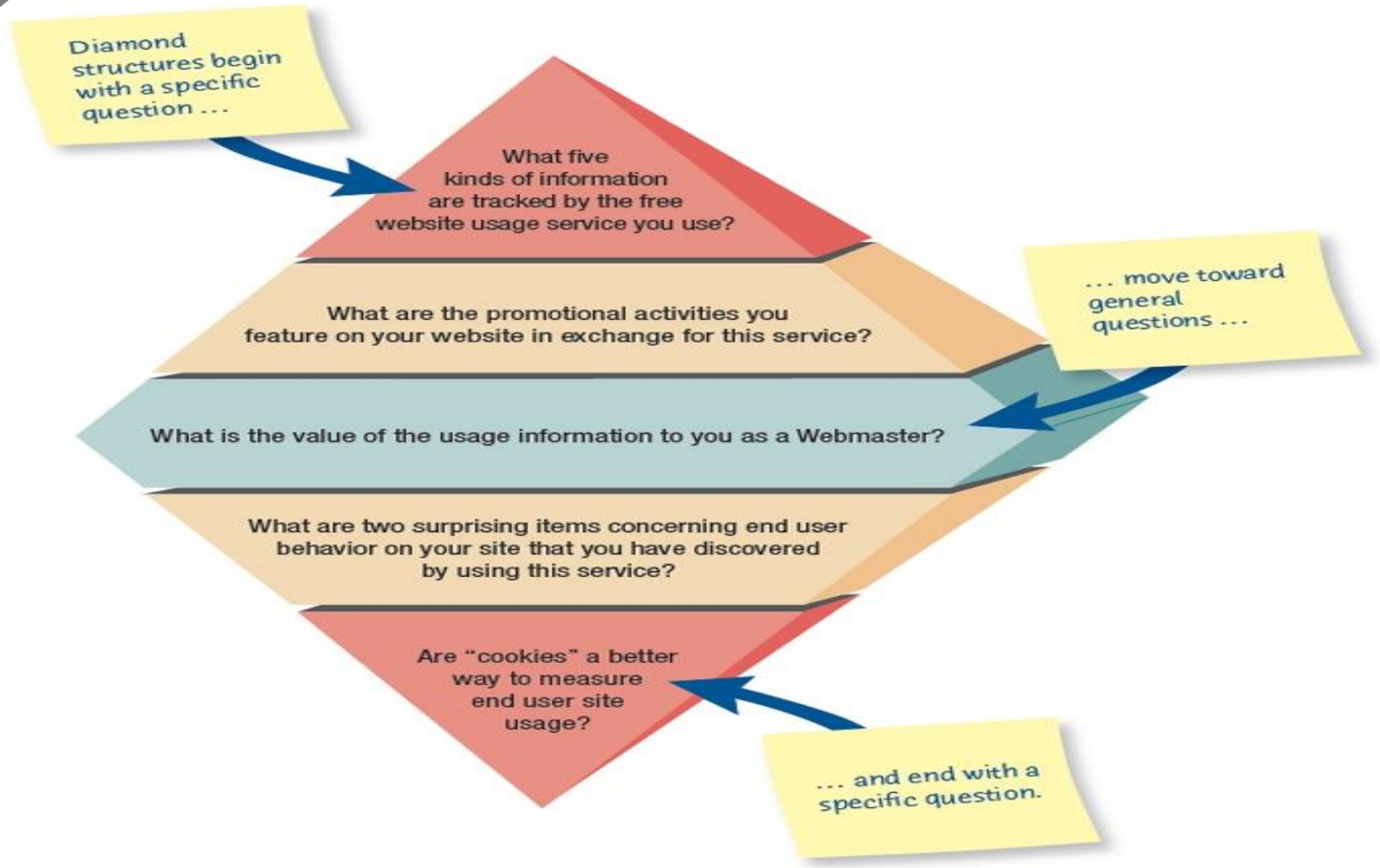
Funnel Structure for Interviewing Begins with Broad Questions Then Funnels to Specific Questions



Diamond Structure

- A diamond-shaped structure begins in a very specific way
- Then more general issues are examined
- Concludes with specific questions
- Combines the strength of both the pyramid and funnel structures
- Takes longer than the other structures

Diamond-Shaped Structure for Interviewing Combines the Pyramid and Funnel Structures



Closing the Interview

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- Always ask “Is there anything else that you would like to add?”
- Summarize and provide feedback on your impressions
- Ask whom you should talk with next
- Set up any future appointments
- Thank them for their time and shake hands.

4.1.4 Interview Report

- Write as soon as possible after the interview
- Provide an initial summary, then more detail
- Review the report with the respondent

4.2 Stories

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- Stories originate in the workplace
- Organizational stories are used to relay some kind of information
- When a story is told and retold over time it takes on a mythic quality
- Isolated stories are good when you are looking for facts
- Enduring stories capture all aspects of the organization and are the ones a systems analyst should look be seeking

Listening to Stories

- Listening to stories is not efficient
- It takes considerably more time than asking interview questions
- Listening to stories may be more rewarding
- Stories are more easily remembered than interview responses

Stories are Made up of Elements

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- All stories have these elements:
  - The call to adventure
  - The quest
  - The struggle
  - The transformation
  - The resolution
  - The moral
  - The epilogue

Reasons for Telling Stories

- **Rich information from listening carefully to the stories is in itself valuable**
- **The information gleaned from the stories will make more sense and be more valuable if seen in context**

Business Stories

- **Business stories can be broken down into four important main types:**
  - **Experiential stories**
  - **Explanatory stories**
  - **Validating stories**
  - **Prescriptive stories**

Stories and the Organization

- Engage organization participants by reacting to stories
- Match one story to another by recounting it to other participants, and collaborating with the stories
- It is a way to deeply understand some of the problems associated with information systems

Purposes for Telling a Story

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- There are four purposes for telling a story:
  - Experiential stories describe what the business or industry is like
  - Explanatory stories tell why the organization acted a certain way
  - Validating stories are used to convince people that the organization made the correct decision
  - Prescriptive stories tell the listener how to act
- Systems analysts can use storytelling as a complement to other information gathering methods

4.3 Joint Application Design (JAD)

- Joint Application Design (JAD) can replace a series of interviews with the user community
- JAD is a technique that allows the analyst to accomplish requirements analysis and design the user interface with the users in a group setting

Conditions That Support the Use of JAD

- Users are restless and want something new
- The organizational culture supports joint problem-solving behaviors
- Analysts forecast an increase in the number of ideas using JAD
- Personnel may be absent from their jobs for the length of time required

Who is Involved

- The people involved are:
  - Executive sponsor
  - IS analyst
  - Users
  - Session leader
  - Observers
  - Scribe

Where to Hold JAD Meetings

- Offsite
  - Comfortable surroundings
  - Minimize distractions
- Attendance
  - Schedule when participants can attend
  - Agenda
  - Orientation meeting

Benefits of JAD

- Time is saved, compared with traditional interviewing
- Rapid development of systems
- Improved user ownership of the system
- Creative idea production is improved

Drawbacks of Using JAD

- JAD requires a large block of time to be available for all session participants
- If preparation or the follow-up report is incomplete, the session may not be successful
- The organizational skills and culture may not be conducive to a JAD session

4.4 Questionnaires

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Questionnaires are useful in gathering information from key organization members about:

- **Attitudes**
- **Beliefs**
- **Behaviors**
- **Characteristics**

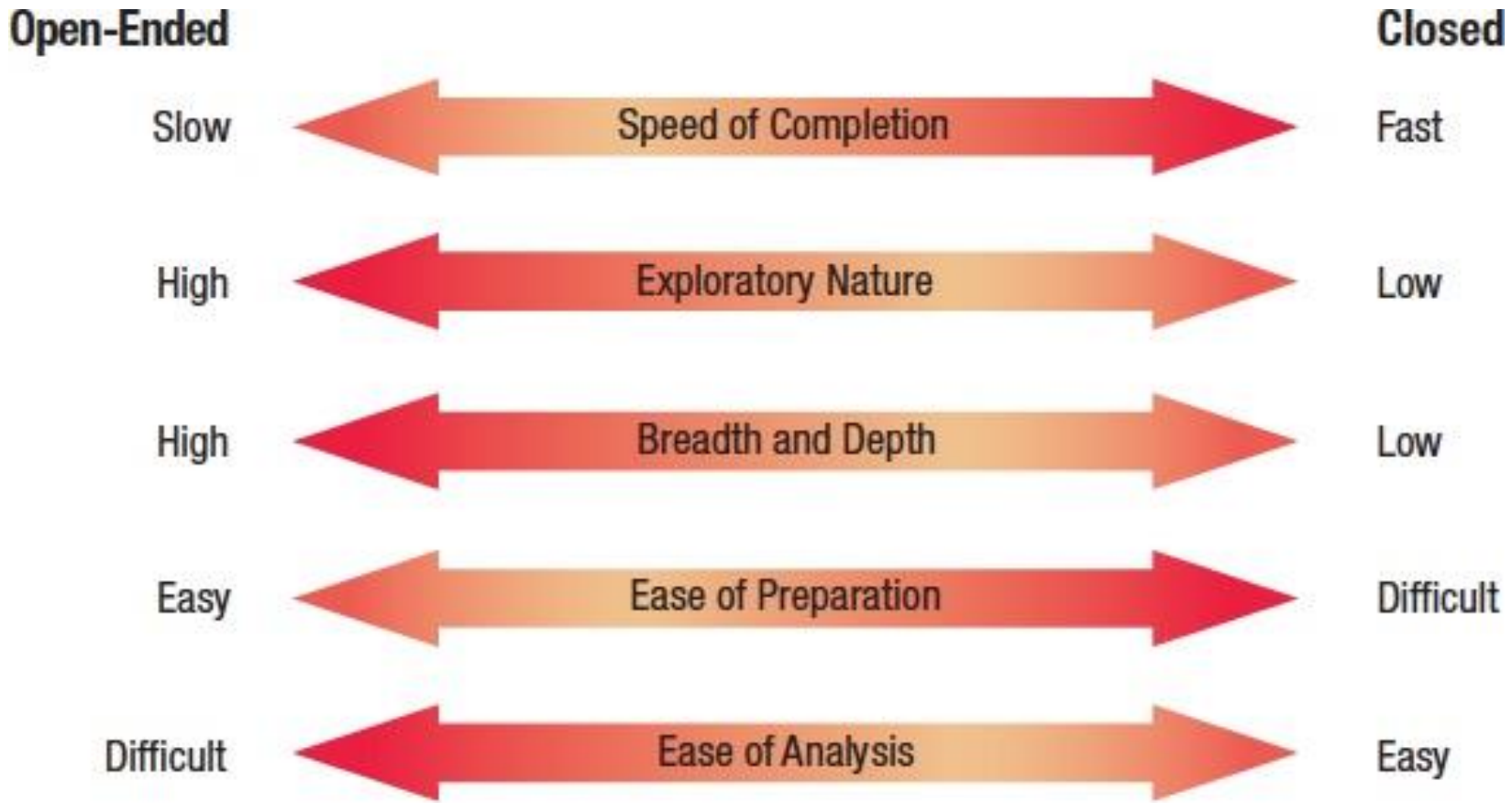
Planning for the Use of Questionnaires

- Organization members are widely dispersed
- Many members are involved with the project
- Exploratory work is needed
- Problem solving prior to interviews is necessary

Question Types (2 of 2)

- Questions are designed as either:
  - Open-ended
    - Try to anticipate the response you will get
    - Well suited for getting opinions
  - Closed
    - Use when all the options may be listed
    - When the options are mutually exclusive

Trade-Offs折中 between the Use of Open-Ended and Closed Questions on Questionnaires



Questionnaire Language

- Simple
- Specific
- Short
- Not patronizing
- Free of bias
- Addressed to those who are knowledgeable
- Technically accurate
- Appropriate for the reading level of the respondent

Measurement Scales

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- The two different forms of measurement scales are:
  - Nominal类别标尺
  - Interval区间标尺

Nominal Scales

- **Nominal scales are used to classify things**
- **It is the weakest form of measurement**
- **Data may be totaled**

What type of software do you use the most?

1 = Word Processor

2 = Spreadsheet

3 = Database

4 = An Email Program

Interval Scales

- **An interval scale is used when the intervals are equal**
- **There is no absolute zero**
- **Examples of interval scales include the Fahrenheit or Centigrade scale**

How useful is the support given by the Technical Support Group?

Not Useful At All				Extremely Useful
1	2	3	4	5

Validity and Reliability

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- Reliability of scales refers to consistency in response—getting the same results if the same questionnaire was administered again under the same conditions
- Validity is the degree to which the question measures what the analyst intends to measure

Problems with Scales

- Leniency（不严格）
- Central tendency（集中趋势）
- Halo effect（光圈效应）

Leniency

- Caused by easy raters
  - Solution is to move the “average” category to the left or right of center

Central Tendency

- Central tendency occurs when respondents rate everything as average
  - Improve by making the differences smaller at the two ends
  - Adjust the strength of the descriptors
  - Create a scale with more points

Halo Effect

- When the impression formed in one question carries into the next question
- Solution is to place one trait and several items on each page

Designing the Questionnaire

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- Allow ample white space
- Allow ample space to write or type in responses
- Make it easy for respondents to clearly mark their answers
- Be consistent in style

Order of Questions

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- Place most important questions first
- Cluster items of similar content together
- Introduce less controversial questions first

Administering Questionnaires

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- Administering questionnaires has two main questions:
  - Who in the organization should receive the questionnaire
  - How should the questionnaire be administered

Ways to Capture Responses When Designing a Web Survey

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Name	Appearance	Purpose
One-line text box		Used to obtain a small amount of text and limit the answer to a few words
Scrolling text box		Used to obtain one or more paragraphs of text
Check box		Used to obtain a yes-no answer (e.g., Do you wish to be included on the mailing list?)
Radio button		Used to obtain a yes-no or true-false answer
Drop-down menu		Used to obtain more consistent results (Respondent is able to choose the appropriate answer from a predetermined list [e.g., a list of state abbreviations])
Push button		Most often used for an action (e.g., a respondent pushes a button marked “Submit” or “Clear”)

Electronically Submitting Questionnaires

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- Reduced costs
- Collecting and storing the results electronically

Summary (1 of 2)

- Interviewing
  - Interview preparation
  - Question types
  - Arranging questions
  - The interview report
- Stories
  - Story elements

Summary (2 of 2)

- Joint Application Design (JAD)
  - Involvement and location
- Questionnaires
  - Writing questions
  - Using scales and overcoming problems
  - Design and order
  - Administering and submitting

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Thank for your Listening.

System Analysis and Design

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